QuantumFlex Security & Data-Governance Statement

Institutional Assurance for Confidential Legal, Financial, and Technical Diligence

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1. Zero-Cloud Data Sovereignty

QuantumFlex processes all client documents, embeddings, and analytical synthesis exclusively on isolated local hardware.

- **No External API Egress:** Client files are never transmitted to public cloud AI providers, third-party model APIs, or external SaaS indexing services.
- **No Model Training on Client Data:** Client materials are never used to train, fine-tune, or update shared foundation models.
- **Loopback-Enforced Execution:** Local inference endpoints are programmatically restricted to loopback interfaces (‘127.0.0.1‘), failing closed if any external network endpoint is configured.

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2. Strict Corpus Isolation & Role Governance

Every document ingested into the QuantumFlex pipeline is assigned an explicit security classification prior to indexing:

- **Client Corpus Isolation:** Each client engagement executes within a dedicated, isolated document store.
- **Boundary Enforcement:** Automated access controls prevent cross-matter retrieval and block internal operational records from entering any client-facing index or report.
- **Idempotent Purge & Retention Control:** Upon engagement completion, client indices can be archived as a deterministic snapshot for client audit recovery or cryptographically purged upon written instruction.

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3. Pre-Ingestion Privacy Sanitization

Before any document page is indexed or analyzed, an automated sanitization filter scans for and redacts sensitive ambient identifiers:

- Unrotated API keys, bearer tokens, and PEM-encoded private keys (‘[REDACTED_CREDENTIAL]‘ / ‘[REDACTED_PRIVATE_KEY]‘).
- Social Security Numbers and personal tax identifiers (‘[REDACTED_SSN]‘).
- Financial routing and bank account numbers (‘[REDACTED_FAMILY_FINANCIAL_ACCT]‘).
- Local workstation filesystem paths.

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4. Deterministic Citation & Tamper Detection

To eliminate unverified generative claims, QuantumFlex enforces a fail-closed cryptographic verification gate (‘AT-RDOS-02‘ / ‘AT-RDOS-03‘):

1. **Chunk-Level SHA-256 Hashing:** Every extracted document segment is hashed with ‘SHA-256‘ at intake.
2. **Mandatory Page & Chunk Binding:** Every analytical claim in a delivered dossier must reference a valid ‘[chunk_id:p<page>]‘ identifier. Uncited claims or page mismatches cause compilation to fail closed.
3. **Corpus Merkle Root:** All verified chunk digests are combined into a deterministic

‘SHA-256‘ Merkle root recorded in the accompanying JSON Admission Receipt. Modifying even a single character in a source document alters the Merkle root and is immediately detected by independent verification.

4. **Honest Assurance Reporting:** Cryptographic receipts explicitly distinguish between locally verified provenance and hardware-signed production attestations (‘signed: false‘ when external hardware ceremony gates are not invoked).